

Research & Development Artefacts

Reliability-Aware Rank Aggregation for Deployment-Oriented Predictive Maintenance Model Selection

Executive Summary

This handover report collates and explains the current state of the SIT747 research project on Reliability-Aware Deployment Rank Aggregation (RDRA). The project investigates deployment-oriented predictive-maintenance model selection when multiple valid deployment criteria produce conflicting candidate rankings and when the empirical evidence supporting those rankings differs in reproducibility. The handover is designed so that future researcher can understand the research direction, identify the final methodology, locate the supporting artefacts, reproduce the principal computational pathway, interpret the current findings correctly, and continue the research without requiring clarification from the original researcher.

Computational work has evolved through six Google Colab notebooks. Notebooks 1-5 preserve the earlier experimental and methodological development, while the Notebook 6 is the authoritative final RDRA artefact. Notebook 6 consumes the primary AI4I evidence generated by Notebook 1 and complementary C-MAPSS evidence generated by Notebook 4, then produces the final reliability estimates, Equal-Borda and RDRA rankings, VQ1-VQ3 validation outputs, exploratory reliability-influence calibration, and complementary cross-task analysis in the `novelty_rank_regret_outputs_revised` folder.

The strongest current evidence is deliberately bounded. Primary RDRA improves several ranking-consistency measures relative to Equal Borda under VQ1 criterion uncertainty. A separately calibrated and locked exploratory refinement ($\gamma = 3$) shows statistically stronger evidence-to-rank responsiveness under the tested VQ2 AI4I corruption scenarios. VQ3 does not demonstrate an additional Top-3 downstream decision-usefulness advantage over Equal Borda. C-MAPSS broadens the evaluation to a sequential RUL setting but is complementary evidence rather than a second complete RDRA replication.

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1. Purpose and Scope

1.1 Purpose of this handover

The purpose of this handover is to preserve the research project as a complete, interpretable and reproducible research package. The package records the development of the research problem, the final RDRA methodology, the computational implementation, source data and provenance, intermediate and final outputs, scientific validation, limitations, publication preparation, and the long-term research direction.

1.2 Scope of the reviewed source set

This report is grounded in the research materials supplied for the current project state. The reviewed source set contains seven core research PDFs, six computational notebooks, and four major output archives.

Artefact class	Reviewed material
Core research documents	Research Progress Check Point 1 Research Progress Check Point 2 Complete Literature Review Complete Research Methodology Research Progress Check Point 3 Research Implementation Research Analysis
Source notebooks	01_AI4I_IPMEF_Final_Updated 02_AI4I_Statistical_Validation_Updated 03_AI4I_Robustness_and_Decision_Extension_Updated 04_CMAPSS_RUL_IPMEF_Updated 05_CMAPSS_Anomaly_Triggered_IPMEF_Updated 06_RDRA_Revised_Methodology_Aligned_Refined(1)
Major output archives	ai4i_ipmef_final_outputs (Notebook 1)

	cmapss_rul_ipmef_outputs (Notebook 4) cmapss_anomaly_triggered_ipmef_outputs (Notebook 5) novelty_rank_regret_outputs_revised (Notebook 6)
Additional finalisation items	GitHub repository URL: https://github.com/OhWenChi/RDRA-Research-Project.git

1.3 Handover interpretation rule

Earlier notebooks and checkpoints are retained as development artefacts rather than rewritten retrospectively to match the final Main Research Question. They document how the project moved from broad deployment-oriented predictive-maintenance evaluation to a single methodological contribution. Hence, the handover distinguishes historical/supporting artefacts from the authoritative final RDRA artefact.

2. Current Research Project State

2.1 Project title and central research problem

The current research project is titled **Reliability-Aware Rank Aggregation for Deployment-Oriented Predictive-Maintenance Model Selection**. It addresses the practical problem that predictive-maintenance deployment suitability cannot be represented by a single predictive-performance metric. Different deployment dimensions can favour different model configurations, and the criterion-specific rankings produced by those dimensions can also differ in empirical reproducibility.

The final study operationalises five deployment dimensions: Predictive Quality, Robustness, Probability Reliability, Computational Efficiency and Operational Impact. These are the five dimensions used in this study, not an exhaustive definition of all industrial deployment requirements.

2.2 Main objective

To develop and empirically validate a reliability-aware rank-aggregation methodology for deployment-oriented predictive-maintenance model selection that accounts for differences in the stability of evaluation evidence and produces a defensible consensus ranking under evaluation uncertainty.

2.3 Main Research Question and Validation Questions

How can reliability-aware rank aggregation produce a robust and defensible consensus ranking of predictive-maintenance model configurations when deployment criteria provide conflicting and differently stable empirical evidence?

Validation question	Purpose
VQ1 - Criterion Uncertainty	Does RDRA provide stronger consensus consistency when the available deployment criteria vary?
VQ2 - Data-Quality Uncertainty	Does the ranking remain appropriately responsive when meaningful changes occur in the underlying deployment evidence?
VQ3 - Preference Uncertainty	Does the resulting consensus support useful downstream deployment decisions when stakeholder priorities vary?

2.4 Proposed methodological contribution and novelty boundary

The contribution is not the invention of Borda scoring, Kendall/Spearman rank agreement, weighted multi-criteria decision making, Pareto analysis, regret analysis, probability calibration or robustness testing. These are established techniques. The proposed adaptation is to estimate criterion-level empirical ranking reproducibility from repeated predictive-maintenance experiments and use that reproducibility as reliability information within deployment-oriented rank aggregation.

An empirical reliability weight reflects the stability of the underlying evidence, whereas a stakeholder-preference weight reflects decision priorities. The two are conceptually distinct and should not be conflated.

This distinction is a defining interpretation rule for the project. A criterion can be operationally important but supported by unstable empirical ranking evidence; conversely, a criterion can be highly reproducible but relatively unimportant to a particular stakeholder. RDRA addresses evidence confidence, while stakeholder preference is treated as a separate downstream decision layer.

3. Research Development History

The project changed direction during the trimester as supervisor feedback and empirical results clarified the strongest defensible contribution. The development history explains why earlier notebooks contain RQ1-RQ3 terminology while the final methodology uses one Main Research Question with three validation questions (VQ1-VQ3).

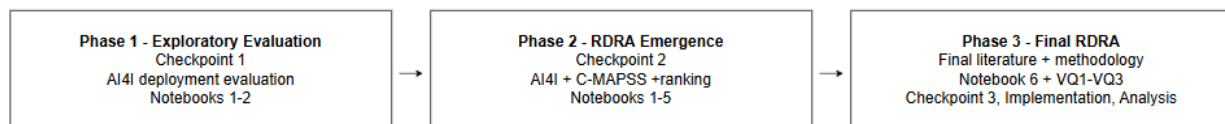


Figure 3.1 Research-development progression from broad deployment evaluation to the final RDRA methodology

3.1 Phase 1 - Exploratory deployment-oriented evaluation

Research Progress Check Point 1 focused on expanding an existing AI4I benchmark beyond single-run predictive performance. The work introduced repeated experimentation, class-imbalance handling, cost-sensitive threshold selection, probability calibration, robustness testing and statistical validation. At this stage, NASA C-MAPSS work had not yet started and the research was still organised around separate questions concerning model comparison, robustness and deployment trade-offs. This phase is principally preserved by Notebooks 1 and 2.

3.2 Phase 2 - Cross-task expansion and RDRA emergence

Research Progress Check Point 2 expanded the experimental work to both AI4I and C-MAPSS. Multiple deployment criteria were found to favour different candidates, which created a model-selection problem that could not be solved simply by adding more metrics. The project then investigated consensus ranking, adopted Equal Borda as a transparent baseline, and developed the initial reliability-aware aggregation idea. Notebooks 1-5 collectively preserve this phase, including robustness extensions, C-MAPSS RUL evaluation and the supplementary anomaly-triggered RUL pipeline.

3.3 Phase 3 - Final RDRA consolidation and validation

By Check Point 3, the research was reorganised around one central method: RDRA. Earlier robustness, sensitivity and downstream decision analyses were repositioned as validation perspectives rather than separate contributions. The final Literature Review and Research Methodology formalised the research gap, Main RQ, VQ1-VQ3, LORO reliability estimation and bounded novelty claim. The updated Notebook 6 is the authoritative implementation of this final research direction, while Research Implementation documents the code-to-method correspondence and Research Analysis interprets the empirical evidence.

4. Handover Repository Architecture

4.1 Recommended repository structure

GitHub Repository Link: <https://github.com/OhWenChi/RDRA-Research-Project.git>

Project Repository Structure

The handover repository is organised into source notebooks, generated experimental outputs, and core research documentation. The root-level README and data/environment files provide instructions for understanding and rebuilding the project.

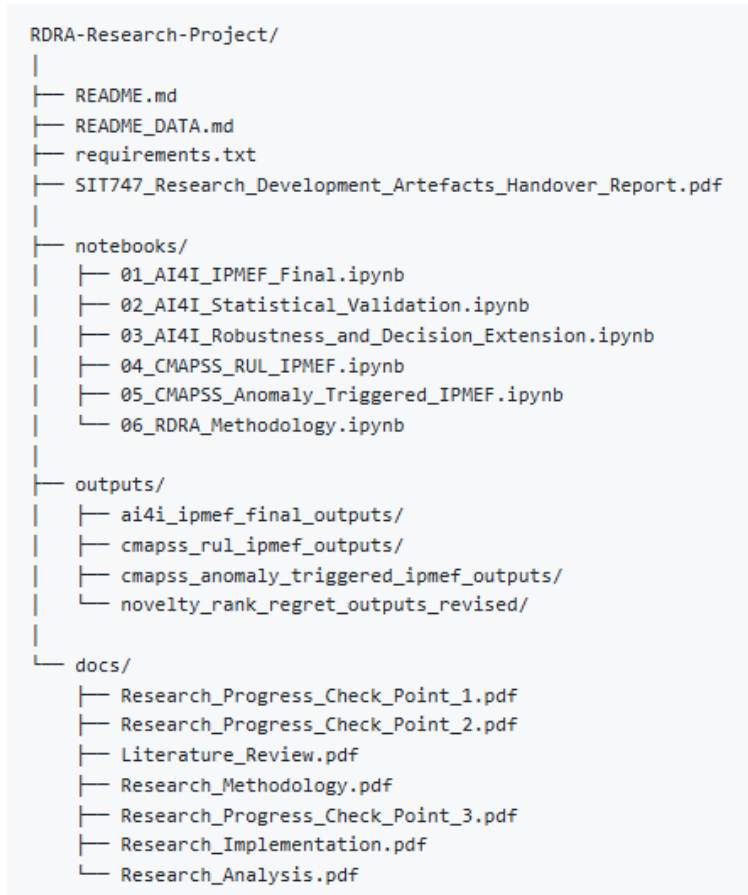


Figure 4.1 Project Repository Structure

The project repository is organised to support reproducibility, traceability and research handover. The notebooks/ directory contains the experimental and analytical pipelines, the outputs/ directory stores generated computational artefacts, and the docs/ directory consolidates the formal research documentation produced throughout the project. The root-level README, data-provenance and dependency files provide additional guidance for understanding and rebuilding the research workflow. This structure enables the project state, computational dependencies and supporting evidence to be reviewed efficiently by supervisors and future researchers.

4.2 Repository and source-version record

The six notebooks constitute the source-code record for the project. Notebooks 1-5 preserve experimental and methodological development, while the Notebook 6 is designated as the authoritative final RDRA implementation. The repository structure separates source notebooks, persisted outputs, research documents, and handover documentation.

4.3 Final computational dependency structure

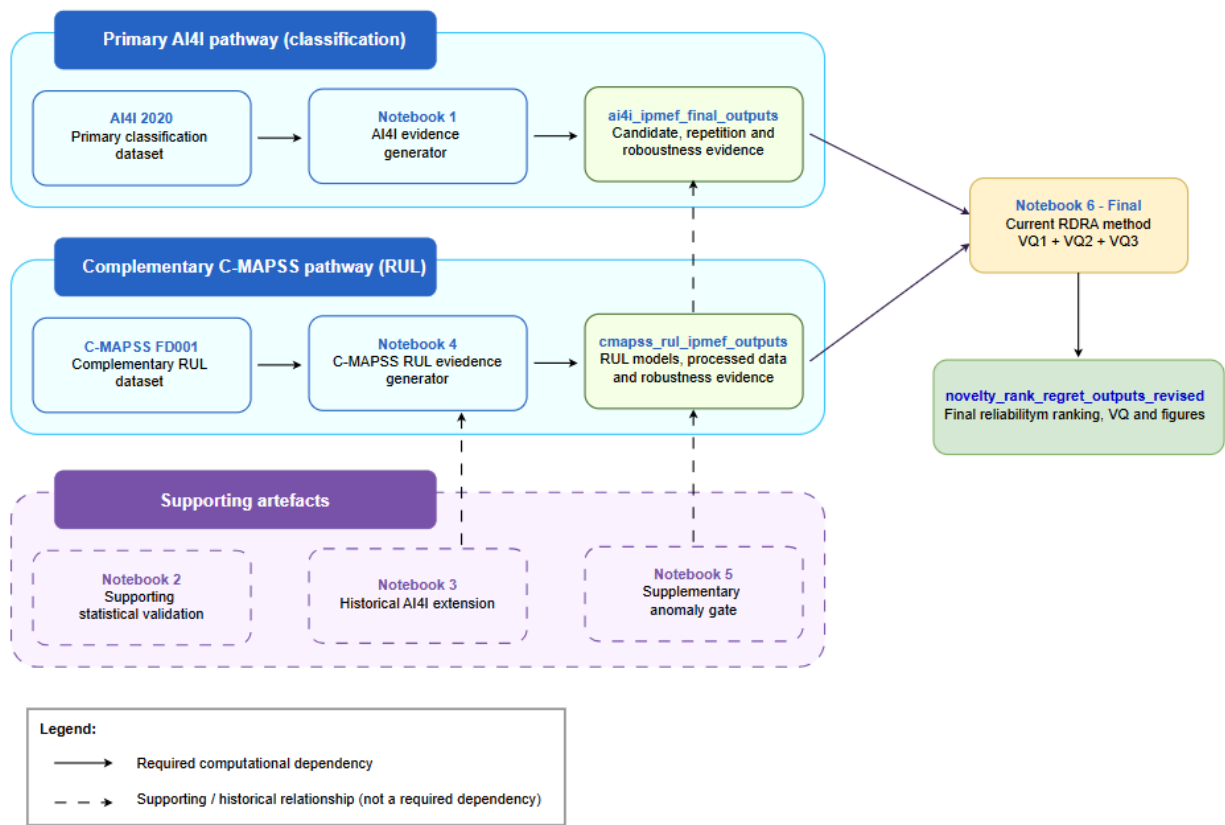


Figure 4.1 . Computational dependency map.

Notebook 6 is the final RDRA artefact.

5. Research Documents and Literature Artefacts

5.1 Core research document inventory

Document	Handover role	Status
Research Progress Check Point 1	Early AI4I development, repeated evaluation, imbalance, calibration, robustness and statistical validation	Historical - Phase 1
Research Progress Check Point 2	AI4I + C-MAPSS expansion; consensus-ranking problem; initial RDRA and ranking-validation direction	Historical/transitional - Phase 2

Complete Literature Review	Final theoretical positioning, research gap, novelty boundary, literature basis for RDRA	Current
Complete Research Methodology	Final objective, Main RQ, VQ1-VQ3, datasets, five dimensions, LORO, RDRA and validation plan	Current/final research design
Research Progress Check Point 3	Methodological consolidation, statistical validation, bounded findings and publication transition	Current - Phase 3
Research Implementation	Functional Python/Google Colab artefact, code-to-equation correspondence and scientific testing	Current/final implementation
Research Analysis	Integrated interpretation of criterion reliability, VQ1-VQ3, C-MAPSS, limitations and final research claim	Current/final analysis

5.2 Literature sources and repositories

The Complete Literature Review is the main human-readable literature artefact. It synthesises predictive-maintenance deployment/model selection, robustness and data-quality degradation, probability calibration, computational/operational constraints, multi-criteria decision making, rank aggregation, experimental uncertainty and ranking reproducibility. It also establishes the novelty boundary: weighted ranking and robust aggregation already exist; the study-specific contribution is the use of repeated-experiment criterion-ranking reproducibility as empirical reliability information in deployment-oriented aggregation.

The Complete Literature Review is accompanied by full scholarly reference lists in the Literature Review and Research Methodology. Therefore, these materials provide the literature-source record used to position the research across predictive maintenance, robustness and data quality, probability calibration, multi-criteria decision making, rank aggregation, uncertainty and empirical ranking reproducibility.

6. Source Code and Artefact Versions

6.1 Notebook inventory and status

Notebook	Function	Status	Dependency/output
01_AI4I_IPMEF_Final_Updated.ipynb	Primary AI4I experimental evidence generator	Essential upstream	Produces ai4i_ipmef_final_outputs
02_AI4I_Statistical_Validation_Updated.ipynb	Controlled ablation and early statistical validation	Supporting/historical	Independent supporting validation
03_AI4I_Robustness_and_Decision_Extension_Updated.ipynb	Earlier corruption sensitivity, constraints, Pareto/TOPSIS and decision extensions	Supporting/historical	Consumes Notebook 1 output
04_CMAPSS_RUL_IPMEF_Updated.ipynb	FD001 RUL evidence generator, models, temporal corruption and processed handoff	Essential upstream complementary	Produces cmapss_rul_ipmef_outputs
05_CMAPSS_Anomaly_Triggered_IPMEF_Updated.ipynb	Anomaly-gated RUL pipeline with Isolation Forest, LSTM autoencoder and hybrid detector	Supplementary/exploratory	Consumes Notebook 4 output
06_RDRA_Revised_Methodology_Aligned_Refined(1).ipynb	Final RDRA, LORO, Equal Borda, VQ1-VQ3, gamma refinement and C-MAPSS cross-task analysis	PRIMARY FINAL ARTEFACT	Consumes Notebook 1 + 4 outputs; produces final RDRA folder

6.2 Source-version status

The notebook set preserves meaningful methodological stages rather than artificial duplicate versions. Notebooks 1-5 retain their historical/supporting roles, and the Notebook 6 is the authoritative implementation of the final Main RQ and VQ1-VQ3 structure. Source history is interpreted through the notebook sequence and the documented research-development phases in Section 3.

7. Data, Models and Experimental Output Artefacts

7.1 AI4I data and primary output archive

Notebook 1 records the AI4I source URL in its experiment configuration and produces the ai4i_ipmef_final_outputs directory. The experiment uses 20 seeds, an asymmetric operational penalty (FP = 1, FN = 10), four corruption families with four severity levels each, five corruption repeats, and repeated latency measurements. The output archive contains 18-candidate summaries, repetition-level clean results, robustness raw evidence, condition-level summaries, deployment rankings, Pareto/scenario artefacts, figures and environment/provenance JSON files.

Key AI4I file	Rows / type	Purpose
01_clean_candidate_results.csv	360 x 30	20 repetitions x 18 candidate configurations; primary repeated evidence
02_robustness_raw_results.csv	23,040 x 14	Raw corruption evidence across candidates, seeds, conditions, severities and repeats
05_composite_robustness_by_seed.csv	360 x 6	Seed-level robustness summary used by final RDRA evidence construction
07_candidate_summary.csv	18 x 29	Candidate-level summary consumed by Notebook 6
experiment_configuration.json	JSON	Seeds, penalty assumptions, corruption levels and data source
execution_environment.json	JSON	Python/platform and core package versions
hyperparameter_provenance.json	JSON	Model and hyperparameter provenance

7.2 C-MAPSS FD001 data and RUL output archive

Notebook 4 works with standard FD001 files (train_FD001.txt, test_FD001.txt and RUL_FD001.txt), creates engine-level train/validation splits, uses a 30-cycle sliding window and a capped RUL

target of 125, and preserves processed sequence arrays, scaling artefacts, metadata and trained models. The current C-MAPSS modelling configuration uses one primary modelling seed (42), which is one reason this branch is complementary rather than a complete independent repeated-evidence RDRA replication.

Key C-MAPSS file	Rows / type	Purpose
01_clean_results.csv	5 x 13	Clean performance/efficiency results for Ridge, RF, XGBoost, LSTM and GRU
02_robustness_raw.csv	500 x 10	Temporal corruption evidence consumed by Notebook 6
06_candidate_summary.csv	5 x 28	Candidate summary consumed by Notebook 6
processed/fd001_primary_sequences.npz	NPZ	Processed sequence tensors for reproducibility
processed/fd001_standard_scaler.joblib	Model artefact	Training-derived standardisation object
models/*.joblib / *.keras	5 trained models	Reproducibility and downstream extension support
processed/notebook5_handoff_manifest.json	JSON	Explicit handoff from Notebook 4 to Notebook 5

7.3 Supplementary anomaly-triggered C-MAPSS archive

Notebook 5 is retained as a supplementary development artefact. It tests anomaly-gated invocation of the downstream RUL predictor using Isolation Forest, an LSTM autoencoder and a hybrid detector, with optional periodic safeguards. Its output folder contains selected detector thresholds, pipeline comparisons, bootstrap intervals, paired Wilcoxon tests, robustness evidence, Pareto/scenario outputs, trigger audits, figures and trained anomaly detectors. Thus, it is provenance/supporting evidence rather than a mandatory final dependency.

7.4 Final RDRA output archive

The novelty_rank_regret_outputs_revised folder is the authoritative output package for Notebook 6. It contains 41 entries including final reliability weights, rank-method comparison, VQ1 detail/summary/statistical files, dominance consistency, VQ2 fixed-clean responsiveness and

method comparison, normalisation sensitivity, VQ3 decision-usefulness outputs, cross-task C-MAPSS responsiveness, gamma-development search, refined VQ1/VQ2 outputs, method reports and final figures. All outputs can be seen in the GitHub repository.

Output group	Representative files	Research role
Reliability and baseline ranking	01_ai4i_dimension_reliability_weights_LOO.csv 02_ai4i_rank_method_comparison_revised.csv	Constructs and reports primary RDRA evidence-confidence weights and candidate rankings
VQ1 criterion uncertainty	03_vq1_criterion_subset_consistency_detail.csv 03b_vq1_criterion_subset_consistency_summary.csv 03d_vq1_rdra_vs_borda_statistical_validation.csv 04_vq1_rdra_leave_one_dimension_out.csv 05_vq1_deployment_dominance_consistency.csv	25 true criterion-subset perturbations, paired statistics, leave-one-dimension-out and dominance checks
VQ2 data-quality uncertainty	06_vq2_ai4i_fixed_clean_responsiveness.csv 06c_vq2_evidence_rank_responsiveness.csv 06d_vq2_rdra_responsiveness_statistical_validation.csv 06g_vq2_rdra_vs_borda_responsiveness_difference.csv 07_vq2_normalisation_sensitivity.csv	Fixed-clean corruption responsiveness, RDRA/Borda comparison and normalisation sensitivity
VQ3 preference uncertainty	08_vq3_pareto_minimax_regret_summary.csv 09_vq3_ranking_shortlist_decision_usefulness.csv 10_vq3_preference_corruption_regret_summary.csv	Pareto/minimax regret, Top-3 shortlist usefulness and joint preference/corruption risk
Cross-task evidence	11_cross_task_cmapss_responsiveness.csv	Complementary deployment-rank response in FD001
Exploratory refinement	12_exploratory_gamma_development_search.csv 13_refined_rdra_vq1_holdout_detail.csv 13b_refined_rdra_vq1_holdout_summary.csv 14_refined_rdra_vq2_responsiveness_raw.csv 14b_refined_rdra_vq2_responsiveness_summary.csv 14c_refined_rdra_vq2_responsiveness_difference.csv	Development-only gamma search, held-out VQ1, locked refined VQ2 and refinement metadata

Figures	figures/vq1_criterion_subset_consistency.png figures/vq1_rdra_vs_borda_95ci.png figures/vq2_ai4i_fixed_clean_responsiveness.png figures/vq2_refined_rdra_vs_equal_borda_responsiveness.png figures/vq2_responsiveness_difference_95ci.png figures/cross_task_cmapss_responsiveness.png	Presentation-ready VQ1, VQ2 and C-MAPSS visual evidence
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8. RDRA Method and Computational Correspondence

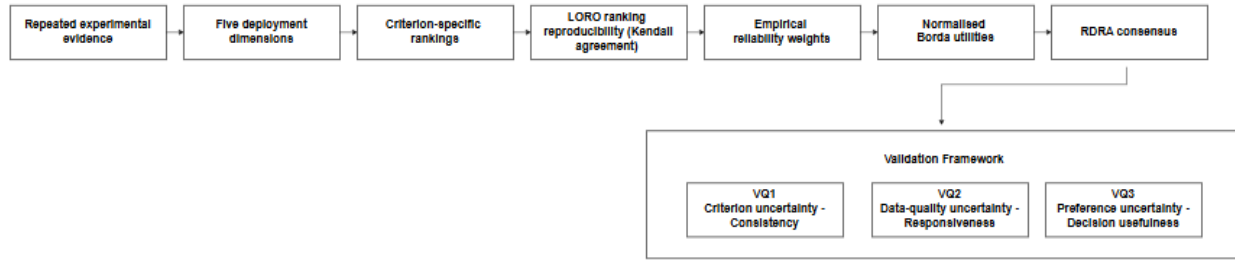


Figure 8.1 Core RDRA method and VQ1-VQ3 validation architecture

8.1 Criterion ranking and Borda transformation

For candidate i and criterion j , r_{ij} denotes the candidate rank, where rank 1 is most desirable. Ties use average ranks to avoid arbitrary ordering. Criterion ranks are transformed to normalised Borda utilities so heterogeneous original units do not need to be added directly.

$$B_{ij} = \frac{n - r_{ij}}{n - 1}$$

8.2 Leave-One-Repetition-Out reliability estimation

For each criterion and each of the 20 AI4I repetitions, one repetition is held out. A reference ranking is constructed from the remaining repetitions, and Kendall agreement is calculated between the held-out and reference candidate rankings. Negative Kendall values are clipped to zero before averaging because contradictory ranking behaviour represents an absence of positive reproducibility evidence rather than a basis for assigning a negative or inverted aggregation weight.

$$s_j = \frac{1}{R} \sum_{r=1}^R \max(0, \tau_{jr})$$

8.3 Empirical reliability weights and consensus

$$w_j = \frac{s_j}{\sum_{k=1}^m s_k}$$
$$S_i = \sum_{j=1}^m w_j B_{ij}$$

Equal Borda uses the same criterion rankings, tie handling, Borda transformation and final score-to-rank pathway but sets all criterion weights to $\frac{1}{m}$. This creates a controlled comparison in which the key methodological difference is empirical reliability-aware criterion influence rather than a different ranking engine.

8.4 Primary versus exploratory RDRA

The supervisor-approved primary RDRA uses $\gamma = 1$, meaning criterion influence is proportional to empirical stability.

A separate exploratory extension uses $w_j(\gamma) \propto s_j^\gamma$. The predefined development grid ranges from $\gamma = 0$ to $\gamma = 3$; $\gamma = 3$ achieved the highest development objective and was locked before subsequent held-out/validation interpretation. Because $\gamma = 3$ is the upper boundary of the tested grid, it is treated as a locked exploratory setting rather than a global optimum.

9. Scientific Evaluation and Current Results

9.1 Empirical reliability weights

Dimension	Positive stability	Reliability weight
Predictive Quality	0.651369	0.222560
Robustness	0.380964	0.130168
Probability Reliability	0.788889	0.269549
Computational Efficiency	0.818301	0.279598
Operational Impact	0.287183	0.098125

The weight pattern supports the motivating premise that the five criterion rankings do not provide equal empirical reproducibility. The weights are protocol-specific evidence-confidence estimates; they are not stakeholder priorities or universal deployment-importance values.

9.2 VQ1 - Criterion uncertainty

Method	Mean Kendall	Mean Spearman	Mean Top-3 overlap	Mean rank displacement
Existing Composite	0.624052	0.741754	0.746667	2.448889
Equal Borda	0.677306	0.799260	0.720000	2.257778
Primary RDRA	0.746405	0.862456	0.813333	1.773333

Across 25 genuine criterion subsets of sizes two, three and four, primary RDRA shows stronger descriptive consistency than Equal Borda on Kendall agreement, Spearman agreement, Top-3 overlap and mean rank displacement. The full five-criterion ranking is used as the reference rather than counted as a 26th perturbation.

Paired VQ1 measure	RDRA improvement	95% CI	Holm-adjusted p	Interpretation
Kendall agreement	+0.069	[0.012, 0.126]	0.039	Statistically supported
Spearman agreement	+0.063	[-0.007, 0.134]	0.077	Directionally positive; not significant after Holm
Top-3 overlap	+0.093	[0.030, 0.156]	0.022	Statistically supported
Rank displacement	0.484 ranks better	[0.146, 0.822]	0.022	Statistically supported

9.3 Additional VQ1 safeguards

The final output package also contains leave-one-dimension-out analysis and a deployment-dominance consistency check. There are nine comparable dominance pairs and zero dominance violations for Existing Composite, Equal Borda and RDRA, indicating that the tested aggregation outputs do not contradict straightforward pairwise dominance relationships in the available evidence.

9.4 Reliability-weight sensitivity

The Research Analysis reports 1,000 moderate perturbations of the empirical reliability-weight vector, with mean Kendall 0.964201, median Kendall 0.973856, mean Top-3 overlap 1.000000, mean absolute rank displacement 0.296444 and tie-aware Top-1 survival 0.699. The central interpretation is that the broader shortlist is substantially more stable than the exact winner. The

current persisted research record retains these aggregated sensitivity results as supporting evidence for the stability of the primary RDRA shortlist.

9.5 VQ2 - Data-quality responsiveness

VQ2 uses 16 AI4I corruption scenarios from Gaussian noise, MCAR missingness, outliers and static sensor offset. A fixed clean-data normalisation reference is retained so corrupted scenarios cannot appear artificially healthy simply because all candidates are re-scaled against one another after deterioration. The analytical relationship is between evidence-change magnitude (ΔE) and resulting rank-change magnitude (ΔR).

Primary $\gamma = 1$ RDRA shows a descriptive responsiveness correlation greater than Equal Borda, but the direct bootstrap interval for the primary difference crosses zero. The stronger inferential finding reported in the final research is therefore based on the separately calibrated and locked exploratory $\gamma = 3$ refinement.

Method	Spearman rho	Difference vs Equal Borda	95% matched-bootstrap CI	Interpretation
Locked refined RDRA (gamma=3)	0.678041	+0.330471	[0.046026, 0.702235]	Statistically stronger responsiveness
Equal Borda	0.347570	-	-	Baseline

9.6 VQ3 - Preference uncertainty

VQ3 applies the same 5,000 Dirichlet stakeholder-preference scenarios to the Top-3 shortlists produced by RDRA and Equal Borda. This is a downstream decision-usefulness test, not a preference-weighted definition of RDRA itself.

Method	Mean regret	P95 regret	CVaR95 regret	Worst-case regret	Zero-regret frequency
Equal Borda	0.002069	0.016802	0.038217	0.079113	0.9316
RDRA	0.002069	0.016802	0.038217	0.079113	0.9316
Existing Composite	0.003082	0.019930	0.057028	0.139577	0.9260

9.7 Complementary C-MAPSS evidence

C-MAPSS FD001 provides cross-task breadth in a sequential RUL setting. The final cross-task output contains 20 condition/severity rows spanning Gaussian noise, gradual offset drift, missing blocks, outliers and sensor freezing. Candidate winners and rank agreement change under these degradations, demonstrating that deployment rankings can respond differently in another

predictive-maintenance task. Because this branch does not estimate a second complete five-dimension reliability-weight vector, it is interpreted as complementary cross-task responsiveness evidence rather than a second full RDRA validation.

9.8 Consolidated current conclusion

Bounded research conclusion

Reliability-aware rank aggregation can produce a more defensible deployment consensus by allowing criterion influence to depend partly on empirical ranking reproducibility rather than assuming identical evidence confidence across all deployment criteria. The strongest demonstrated evidence concerns criterion uncertainty and, for the locked exploratory refinement, data-quality responsiveness. The current evidence does not establish universal downstream superiority.

10. Rebuilding and Reproducing the Project

10.1 Computational environment

The notebooks were developed in Google Colab on Linux with Python 3.12.13. The saved environment records show NumPy 2.0.2 and pandas 2.2.2 for the AI4I and C-MAPSS runs, and TensorFlow 2.20.0 for the C-MAPSS pipelines. Additional packages used across the notebooks are summarised in the table below, while the persisted execution_environment.json files provide run-specific environment provenance.

Package group	Packages used across notebooks
Core numerical/data	numpy, pandas, scipy
Machine learning	scikit-learn, xgboost, imbalanced-learn
Deep learning	tensorflow / keras
Statistics	statsmodels, scipy.stats
Explainability	shap
Persistence/system	joblib, psutil
Visualisation	matplotlib

10.2 Required datasets and provenance

AI4I is retrieved by Notebook 1 from the UCI source recorded in experiment_configuration.json. C-MAPSS FD001 uses the standard train_FD001.txt, test_FD001.txt and RUL_FD001.txt files. The rebuild procedure therefore treats AI4I as programmatically retrievable source data and C-MAPSS FD001 as an externally supplied raw-data dependency with explicitly defined required filenames.

10.3 Final reproduction sequence

1. Open a clean Google Colab/Python 3.12 environment and install the packages listed in Table 10.1 or equivalent compatible versions recorded in the persisted environment files.
2. Run Notebook 1 to generate ai4i_ipmef_final_outputs. Confirm that 07_candidate_summary.csv contains 18 candidates and that 01_clean_candidate_results.csv contains 360 rows (18 candidates x 20 repetitions).
3. Prepare the C-MAPSS FD001 raw files and run Notebook 4. Confirm that 06_candidate_summary.csv contains five candidate models and that processed sequence/scaler artefacts are produced.
4. Place ai4i_ipmef_final_outputs/ and cmapss_rul_ipmef_outputs/ in the working directory expected by the final Notebook 6.
5. Run the updated final Notebook 6. It creates novelty_rank_regret_outputs_revised/.
6. Compare regenerated key outputs with the reference files supplied in the repository, particularly the reliability weights, VQ1 summary/statistical tables, VQ3 Top-3 table, gamma search and refined VQ2 bootstrap difference.
7. Run Notebooks 2, 3 and 5 only when their supporting historical/supplementary analyses are required. They are not mandatory dependencies of the final Notebook 6 results.

10.4 Expected verification checkpoints

Checkpoint	Expected reference
AI4I candidate summary	18 candidate configurations
AI4I repeated clean evidence	360 rows = 18 candidates x 20 repetitions
Final reliability weights	Predictive 0.222560; Robustness 0.130168; Probability Reliability 0.269549; Efficiency 0.279598; Operational 0.098125
VQ1 perturbations	25 true criterion subsets
VQ1 primary RDRA mean Kendall	0.746405
VQ3 preference scenarios	5,000 matched scenarios
Exploratory gamma selection	$\gamma = 3$, selected on development-only VQ1 data
Locked refined VQ2 difference	$\Delta p = 0.330471$; 95% CI [0.046026, 0.702235]

10.5 Reproducibility safeguards

- Preserve the final analytical RNG_SEED = 20260811 used for randomised downstream analyses.
- Do not change the corruption definitions or clean-data normalisation and still claim exact replication of the current VQ2 evidence.
- Do not use VQ2/VQ3 results to re-select gamma. The refinement was calibrated on development-only VQ1 evidence and then locked.

- Document package/library changes if exact numerical replication differs because of environment evolution.
- Keep the primary $\gamma = 1$ RDRA and exploratory $\gamma = 3$ refinement separate in filenames, figures and interpretation.

11. Project Planning and Management

11.1 Checkpoints as project-management artefacts

Checkpoints 1-3 form the project-management record for the research. Therefore, these document milestone progression, impediments, supervisor feedback and corrective actions, including the transition from broad deployment evaluation to the final RDRA-centred methodology. Their retention explains why earlier analyses remain part of the handover and why the final research claim is narrower than the initial project direction.

11.2 Key methodological decisions to preserve

Decision	Reason / handover interpretation
Use Equal Borda as principal controlled baseline	Keeps candidate set, criterion rankings, Borda transform and final ranking pathway common so the main difference is empirical reliability influence.
Treat reliability as evidence reproducibility, not preference	Prevents stakeholder importance from being confused with the strength/repeatability of experimental ranking evidence.
Use LORO Kendall agreement	Measures criterion-specific ordering reproducibility across repeated experiments rather than generalisation error.
Clip negative Kendall values to zero for positive stability	Disagreement reduces evidence confidence but does not reverse the semantic direction of a criterion.
Use fixed clean-data normalisation in primary VQ2 protocol	Prevents corrupted scenarios from being re-scaled into appearing healthy.
Keep primary $\gamma = 1$ separate from exploratory $\gamma = 3$	Avoids retrospectively redefining the proposed method after observing later validation data.
Treat C-MAPSS as complementary evidence	Current C-MAPSS branch lacks the same repeated five-dimension structure required for a full independent reliability-weight vector.
Report neutral VQ3 result	Maintains falsifiability and avoids claiming universal superiority not supported by downstream evidence.

12. Limitations, Risks, Ethics and Sustainability

12.1 Scientific and technical limitations

- C-MAPSS provides complementary three-dimension cross-task evidence rather than a second full RDRA replication.
- VQ1 is constrained by the 25 genuine criterion subsets available from the five-dimension design.
- VQ2 uses 16 controlled AI4I corruption scenarios; other sensor/data failures may produce different ranking behaviour.
- VQ3 does not demonstrate an additional Top-3 decision-usefulness advantage over Equal Borda under the tested preferences.
- $\gamma = 3$ is the upper boundary of the predefined development grid and is not a proven global optimum.
- The study does not yet provide prospective industrial validation with real maintenance costs, safety outcomes or organisational deployment constraints.

12.2 Ethical, safety and responsible-use considerations

The current work uses public benchmark data and does not involve direct human participants or personal information. Nevertheless, predictive-maintenance recommendations can influence operational maintenance decisions, downtime and potentially safety-critical systems. Hence, the framework is positioned as decision-support research rather than an autonomous industrial decision authority. Analytical operational penalties are modelling assumptions, not measured financial or safety costs, industrial deployment would require domain-specific validation, governance and appropriate human oversight.

12.3 Sustainability considerations

Computational Efficiency is explicitly represented as one deployment dimension through inference latency and model size. This recognises that deployment suitability includes resource demands as well as predictive performance. The preserved processed data, trained models and reference outputs also support computational sustainability by allowing future researchers to inspect and extend the work without automatically repeating every expensive experiment. Computational cost remains an explicit consideration for any later deep-learning extension.

13. Long-Term Vision and Future Research Roadmap

The long-term goal is to develop RDRA from a benchmark-based research prototype into a general evidence-confidence-aware model-selection framework for industrial predictive maintenance. The defining research principle is the separation of empirical evidence reliability from stakeholder preference.

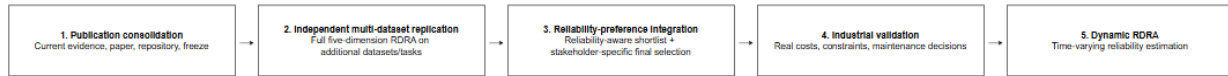


Figure 13.1 Long-term research roadmap

13.1 Priority 1 - Publication consolidation and reproducible dissemination

The immediate research direction is the consolidation of the current Literature Review, Methodology, Implementation and Analysis into a publication-oriented manuscript with traceable figures, tables and computational evidence. Reproducible dissemination preserves the bounded findings, including the neutral VQ3 result, as part of the scientific record.

13.2 Priority 2 - Full independent RDRA replication

The most important scientific extension is a second dataset/task that supports repeated construction of all five (or an explicitly revised set of) deployment dimensions. This would allow independent estimation of a second reliability-weight vector and direct study of whether criterion reproducibility patterns generalise across tasks.

13.3 Priority 3 - Reliability and stakeholder preference as two explicit layers

A future two-layer decision architecture separates evidence confidence from stakeholder preference. Stage 1 uses reliability-aware aggregation to construct a defensible shortlist; Stage 2 applies stakeholder-specific priorities within that shortlist. This extends VQ3 while retaining the interpretation of RDRA weights as evidence-reproducibility weights rather than importance weights.

13.4 Priority 4 - Stronger reliability-influence calibration

A broader investigation of the gamma reliability-influence parameter is a future methodological direction. Appropriate studies include predefined wider search ranges, nested development/validation, regularisation and Bayesian estimation. The current $\gamma = 3$ result remains exploratory because it lies at the upper boundary of the development grid and has not been established as a global optimum.

13.5 Priority 5 - Industrial and online RDRA

Longer-term validation extends the framework to real industrial maintenance data, measured asymmetric consequences, hardware/latency constraints and changing evidence quality. A dynamic formulation could estimate $s_j(t)$ and $w_j(t)$ over time so that aggregation influence adapts as criterion reliability changes under drift, sensor ageing or new operating conditions.

14. Final Repository Completion and Handover Checklist

14.1 Core research and computational artefacts

- All six computational notebooks, including the updated/final Notebook 6
- Primary AI4I experimental output archive
- C-MAPSS RUL output archive with trained models and processed handoff artefacts
- Supplementary anomaly-triggered C-MAPSS output archive
- Final Notebook 6 novelty_rank_regret_outputs_revised archive
- Research Progress Check Points 1, 2 and 3
- Complete Literature Review
- Complete Research Methodology
- Research Implementation
- Research Analysis

14.2 Handover usability

The package is structured so that a new researcher can determine the final research question, identify the authoritative notebook, trace Notebook 6 dependencies, locate raw and processed data, identify the outputs supporting VQ1-VQ3, distinguish primary from exploratory RDRA, understand the complementary role of C-MAPSS, identify statistically supported and neutral findings, recognise current limitations and follow the documented future research direction without additional methodological clarification.

15. Conclusion

The current project has progressed from broad deployment-oriented predictive-maintenance evaluation to a clearly defined and implemented rank-aggregation research contribution. The final RDRA artefact introduces an evidence-confidence layer based on criterion-ranking reproducibility, evaluates the method under three complementary uncertainty perspectives, and preserves important boundaries around the claims. The source notebooks, experimental output folders and research documents now provide a strong basis for a reproducible SIT747 handover.

Therefore, the source notebooks, persisted output folders, research documents, rebuild procedure, artefact inventory, interpretation safeguards, long-term roadmap and learning-outcome mapping constitute a complete handover description of the current project state. The package provides a reproducible baseline from which supervisors or future researchers can rebuild, interpret and extend the work while preserving the boundaries of the current empirical evidence.

Appendix A. Core Artefact Inventory

ID	Artefact	Format	Role	Status
A01	Research Progress Check Point 1	PDF	Phase 1 historical research progress	Available
A02	Research Progress Check Point 2	PDF	Phase 2 transition to RDRA	Available
A03	Complete Literature Review	PDF	Literature synthesis, gap and novelty positioning	Available
A04	Complete Research Methodology	PDF	Final research design and RDRA methodology	Available
A05	Research Progress Check Point 3	PDF	Phase 3 final validation progress	Available
A06	Research Implementation	PDF	Final computational implementation and testing	Available
A07	Research Analysis	PDF	Final interpretation and research conclusions	Available
C01	Notebook 1 – AI4I IPMEF	IPYNB	Primary AI4I experimental evidence generator	Available
C02	Notebook 2 – AI4I Statistical Validation	IPYNB	Supporting historical statistical validation	Available
C03	Notebook 3 – AI4I Robustness/Decision Extension	IPYNB	Supporting historical decision extension	Available
C04	Notebook 4 – C-MAPSS RUL	IPYNB	Complementary RUL evidence generator	Available
C05	Notebook 5 – C-MAPSS Anomaly Triggered	IPYNB	Supplementary anomaly-gated pipeline	Available
C06	Notebook 6 - Final RDRA	IPYNB	Primary final computational artefact	Available
O01	ai4i_ipmef_final_outputs	Folder /ZIP	Primary AI4I data/results provenance	Available

O02	cmapss_rul_ipmef_outputs	Folder /ZIP	C-MAPSS data/models/results provenance	Available
O03	cmapss_anomaly_triggered_ipmef_outputs	Folder /ZIP	Supplementary anomaly-triggered outputs	Available
O04	novelty_rank_regret_outputs_revised	Folder /ZIP	Final RDRA reliability/VQ/figure evidence	Available
P01	Research Papers Review	PDF	Publication- oriented SIT747 artefact	Completed
P02	Research paper draft	PDF	Current publication manuscript	Completed

Appendix B. Major Output File Manifest

The table below summarises the persisted files in the four reviewed output archives and provides a human-readable file manifest linking the stored artefacts to their analytical roles.

ai4i_ipmef_final_outputs (Notebook 1)

File	Purpose category
01_clean_candidate_results.csv	Candidate / experimental results
02_robustness_raw_results.csv	Robustness evidence
03_robustness_by_level.csv	Robustness evidence
04_condition_auretc_by_seed.csv	Analysis table
05_composite_robustness_by_seed.csv	Robustness evidence
06_composite_robustness_summary.csv	Robustness evidence
07_candidate_summary.csv	Candidate / experimental results
08_rq1_conventional_vs_deployment_ranking.csv	Analysis table
09_rq1_summary.csv	Analysis table
10_rq3_pareto_candidates.csv	VQ3 / decision evidence
11_rq3_preliminary_fixed_weight_baseline.csv	Reliability / weighting evidence
12_framework_report.json	Configuration / environment / method report
execution_environment.json	Configuration / environment / method report
experiment_configuration.json	Configuration / environment / method report
figures/explainability_shap_summary.png	Figure / visual evidence
figures/rq1_ranking_comparison.png	Figure / visual evidence
figures/rq2_gaussian_noise_retention.png	Figure / visual evidence

figures/rq2_missing_values_mcar_retention.png	Figure / visual evidence
figures/rq2_outliers_retention.png	Figure / visual evidence
figures/rq2_overall_robustness_heatmap.png	Figure / visual evidence
figures/rq2_static_sensor_offset_retention.png	Figure / visual evidence
figures/rq3_pareto_f1_vs_cost.png	Figure / visual evidence
figures/rq3_pareto_robustness_vs_latency.png	Figure / visual evidence
hyperparameter_provenance.json	Configuration / environment / method report

cmapss_rul_ipmef_outputs (Notebook 4)

File	Purpose category
01b_test_predictions_and_residuals.csv	Analysis table
01_clean_results.csv	Candidate / experimental results
02_robustness_raw.csv	Robustness evidence
03_robustness_by_level.csv	Robustness evidence
04_condition_robustness.csv	Robustness evidence
05_composite_robustness.csv	Robustness evidence
06_candidate_summary.csv	Candidate / experimental results
07_rq1_summary.csv	Analysis table
08_candidate_pareto_summary.csv	VQ3 / decision evidence
09_scenario_recommendations.csv	VQ3 / decision evidence
10_framework_report.json	Configuration / environment / method report
execution_environment.json	Configuration / environment / method report
experiment_configuration.json	Configuration / environment / method report
figures/rq1_best_model_true_vs_predicted_rul.png	Figure / visual evidence
figures/rq1_ranking_comparison.png	Figure / visual evidence
figures/rq2_gaussian_noise_retention.png	Figure / visual evidence
figures/rq2_gradual_offset_drift_retention.png	Figure / visual evidence
figures/rq2_missing_block_retention.png	Figure / visual evidence
figures/rq2_outliers_retention.png	Figure / visual evidence
figures/rq2_overall_robustness_heatmap.png	Figure / visual evidence
figures/rq2_sensor_freezing_retention.png	Figure / visual evidence
models/GRU.keras	Persisted model artefact
models/LSTM.keras	Persisted model artefact
models/Random_Forest.joblib	Persisted model artefact
models/Ridge.joblib	Persisted model artefact
models/XGBoost.joblib	Persisted model artefact
processed/fd001_features.json	Processed data / handoff artefact

processed/fd001_primary_sequences.npz	Processed data / handoff artefact
processed/fd001_standard_scaler.joblib	Persisted model artefact
processed/fd001_test_window_metadata.csv	Processed data / handoff artefact
processed/fd001_train_window_metadata.csv	Processed data / handoff artefact
processed/fd001_validation_window_metadata.csv	Processed data / handoff artefact
processed/notebook5_handoff_manifest.json	Processed data / handoff artefact

cmapss_anomaly_triggered_ipmef_outputs (Notebook 5)

File	Purpose category
01_selected_detector_thresholds.csv	Anomaly-triggered evidence pipeline
02_validation_trajectory_coverage.csv	Analysis table
03_pipeline_comparison.csv	Anomaly-triggered evidence pipeline
04_pipeline_bootstrap_confidence_intervals.csv	Anomaly-triggered evidence pipeline
05_pipeline_paired_wilcoxon_tests.csv	Anomaly-triggered evidence pipeline
06_detector_robustness_raw.csv	Robustness evidence
07_detector_robustness_by_level.csv	Robustness evidence
08_detector_condition_robustness.csv	Robustness evidence
09_detector_composite_robustness.csv	Robustness evidence
10_pipeline_pareto_summary.csv	VQ3 / decision evidence
11b_scenario_rankings.csv	VQ3 / decision evidence
11_scenario_recommendations.csv	VQ3 / decision evidence
12_trigger_audit_table.csv	Anomaly-triggered evidence pipeline
13_framework_report.json	Configuration / environment / method report
execution_environment.json	Configuration / environment / method report
experiment_configuration.json	Configuration / environment / method report
figures/detector_scores_vs_true_rul.png	Figure / visual evidence
figures/rq2_detector_gaussian_noise_coverage.png	Figure / visual evidence
figures/rq2_detector_gradual_offset_drift_coverage.png	Figure / visual evidence
figures/rq2_detector_missing_block_coverage.png	Figure / visual evidence
figures/rq2_detector_outliers_coverage.png	Figure / visual evidence
figures/rq2_detector_sensor_freezing_coverage.png	Figure / visual evidence
figures/rq3_invocation_reduction_vs_missed_critical.png	Figure / visual evidence
models/isolation_forest.joblib	Persisted model artefact

models/lstm_autoencoder.keras	Persisted model artefact
threshold_search_hybrid_detector.csv	Anomaly-triggered pipeline evidence
threshold_search_isolation_forest.csv	Anomaly-triggered pipeline evidence
threshold_search_lstm_autoencoder.csv	Anomaly-triggered pipeline evidence

novelty_rank_regret_outputs_revised (Notebook 6)

File	Purpose category
01_ai4i_dimension_reliability_weights_LOO.csv	Reliability / weighting evidence
02_ai4i_rank_method_comparison_revised.csv	Analysis table
03b_vq1_criterion_subset_consistency_summary.csv	VQ1 / criterion-uncertainty evidence
03c_vq1_consistency_by_subset_size.csv	VQ1 / criterion-uncertainty evidence
03d_vq1_rdra_vs_borda_statistical_validation.csv	VQ1 / criterion-uncertainty evidence
03e_vq1_rdra_vs_borda_paired_differences.csv	VQ1 / criterion-uncertainty evidence
03f_vq1_statistical_summary_for_presentation.csv	VQ1 / criterion-uncertainty evidence
03_vq1_criterion_subset_consistency_detail.csv	VQ1 / criterion-uncertainty evidence
04_vq1_rdra_leave_one_dimension_out.csv	VQ1 / criterion-uncertainty evidence
05b_Equal_Borda_dominance_violations.csv	VQ1 / criterion-uncertainty evidence
05b_Existing_Composite_dominance_violations.csv	VQ1 / criterion-uncertainty evidence
05b_RDRA_dominance_violations.csv	VQ1 / criterion-uncertainty evidence
05_vq1_deployment_dominance_consistency.csv	VQ1 / criterion-uncertainty evidence
06b_vq2_ai4i_condition_summary.csv	VQ2 / data-quality evidence
06c_vq2_evidence_rank_responsiveness.csv	VQ2 / data-quality evidence
06d_vq2_rdra_responsiveness_statistical_validation.csv	VQ2 / data-quality evidence
06e_vq2_rdra_equal_borda_responsiveness_raw.csv	VQ2 / data-quality evidence
06f_vq2_method_responsiveness_summary.csv	VQ2 / data-quality evidence
06g_vq2_rdra_vs_borda_responsiveness_difference.csv	VQ2 / data-quality evidence

06_vq2_ai4i_fixed_clean_responsiveness.csv	VQ2 / data-quality evidence
07_vq2_normalisation_sensitivity.csv	VQ2 / data-quality evidence
08_vq3_pareto_minimax_regret_summary.csv	VQ3 / decision evidence
09_vq3_ranking_shortlist_decision_usefulness.csv	VQ3 / decision evidence
10_vq3_preference_corruption_regret_summary.csv	VQ3 / decision evidence
11_cross_task_cmapss_responsiveness.csv	VQ2 / data-quality evidence
12_exploratory_gamma_development_search.csv	Analysis table
12_revised_rdra_method_report.json	Configuration / environment / method report
13b_refined_rdra_vq1_holdout_summary.csv	VQ1 / criterion-uncertainty evidence
13_refined_rdra_vq1_holdout_detail.csv	VQ1 / criterion-uncertainty evidence
14b_refined_rdra_vq2_responsiveness_summary.csv	VQ2 / data-quality evidence
14c_refined_rdra_vq2_responsiveness_difference.csv	VQ2 / data-quality evidence
14_refined_rdra_vq2_responsiveness_raw.csv	VQ2 / data-quality evidence
16_rdra_refinement_report.json	Configuration / environment / method report
figures/cross_task_cmapss_responsiveness.png	Figure / visual evidence
figures/vq1_criterion_subset_consistency.png	Figure / visual evidence
figures/vq1_rdra_vs_borda_95ci.png	Figure / visual evidence
figures/vq2_ai4i_fixed_clean_responsiveness.png	Figure / visual evidence
figures/vq2_refined_rdra_vs_equal_borda_responsiveness.png	Figure / visual evidence
figures/vq2_responsiveness_difference_95ci.png	Figure / visual evidence

Internal project artefacts used to construct this handover report

Core PDFs: Literature Review; Research Methodology; Research Progress Check Point 3; Research Implementation; Research Analysis.

Core Source notebooks: 01_AI4I_IPMEF_Final; 04_CMAPSS_RUL_IPMEF; 05_CMAPSS_Anomaly_Triggered_IPMEF; 06_RDRA_Methodology.

Output archives: ai4i_ipmef_final_outputs; cmapss_rul_ipmef_outputs; cmapss_anomaly_triggered_ipmef_outputs; novelty_rank_regret_outputs_revised.